Multi-Dimensional Characterization of LLM-Generated Creative Ideas Across Indonesian Design Thinking Domains

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***Abstract*— Large Language Models (LLMs) are increasingly used to generate ideas during the ideation phase of design thinking, yet how such open-ended output should be evaluated remains an open problem, particularly for medium-resource languages such as Indonesian. This study presents a controlled multi-dimensional characterization of creative ideas produced by two locally deployed LLMs of comparable scale, Sahabat-AI and Qwen3-8B. From 136 Indonesian design thinking problem statements across six application domains, five ideas were generated per model per statement, yielding 1,360 ideas evaluated on within-batch lexical distinctiveness, problem-idea relevance, and idea-rationale semantic coherence. The models exhibited opposing profiles: Sahabat-AI scored higher on lexical distinctiveness (rank-biserial +0.592), Qwen3-8B on cosine-based relevance and coherence (-0.539 and -0.659). Cross-domain consistency was dimension-dependent, with the lexically based dimensions preserving their direction across all six domains and cosine-based relevance not. The metric-level analysis revealed three issues: aggregate scores can reverse the direction of the conclusion depending on the weighting chosen, embedding-based measures show sharply concentrated distributions with skewness reaching -3.74, and controlling for output length removes the difference in cosine-based relevance while reducing the coherence difference by about seventy percent. Model comparisons on ideation tasks should therefore report dimensions separately, test sensitivity to aggregation parameters, and control surface variables before inferring differences in capability.**

Keywords*— large language models, creative ideation, design thinking, Indonesian NLP, automatic evaluation metrics

I. Introduction

Ideation is central to design thinking: it is the phase in which the solution space is opened as widely as possible before selection and testing take place [1]. LLMs have changed how the phase is carried out, since alternatives can now be generated quickly, at scale, repeatedly, and reviews of design practice report that the presence of AI is shifting the role and workflow of designers [2], [3]. Evidence on the effect is mixed. One study found no qualitative difference between human ideas and those produced by generative chatbots [4], while other work reports that generative support raises assessments of individual quality but lowers collective diversity [5] and reduces the variety and originality of the output [6].

How that output should be evaluated is the open problem. A creative idea is not a single construct; several characteristics can be distinguished operationally [7], [8]. Lexical distinctiveness within a batch can be operationalized through word-set overlap across outputs [9], whereas problem-idea relevance and idea-rationale coherence are typically measured through embedding-based semantic similarity [10]. A single aggregated score captures only part of these characteristics [8], and the trade-offs are substantive: ideas rated more novel may be rated weaker on feasibility [11], Pareto-front mapping across dozens of models shows originality bought at the cost of feasibility and the reverse [12], and output that appears original individually may show variability at the population level [13].

The problem sharpens for Indonesia, where NLP resources have developed more slowly and LLM capabilities are still assessed largely with English-language datasets [14], [15]. Three gaps follow. First, the available Indonesian generation benchmarks target reference-based tasks [16], so automatic characterization of several output dimensions at once on open-ended tasks remains limited. Second, domain-dependent variation in relative model advantage has been shown for English scientific domains [12] but not for Indonesian design thinking domains, while experimental studies of LLM-assisted ideation generally rest on a single task domain [6]. Third, the metrics carry distributional characteristics of their own: Transformer representations are anisotropic [17], cosine similarity can reflect factors other than semantic proximity [18], and automatic scoring on creative tasks yields a compressed score range [12].

This study therefore compares two locally deployed LLMs of comparable scale, Sahabat-AI [19] and Qwen3-8B [20], under uniform conditions. Three questions guide the analysis: RQ1: how the two models differ on the three dimensions, RQ2: how far those differences hold across domains, and RQ3: what distributional characteristics and limitations the automatic metrics themselves introduce. The paper contributes a controlled cross-domain experimental setting for Indonesian creative ideation, an empirical characterization of two contrasting model profiles, and a metric-level analysis showing how aggregation and output length can each conceal or produce apparent differences in generative behavior.

II. Related Work

A. LLMs for Creative Ideation

Assessing ideation on several dimensions at once has a long precedent in design research, where novelty, variety, quality, and quantity are measured separately rather than collapsed into a single score [21]. Research on LLMs in ideation divides according to what is evaluated. Process-oriented work places the model within a human workflow, as in the integration of LLMs into Brainwriting [2], the examination of how generative systems affect design fixation [6], and a systematic review of the shifting role of the designer [3]. Product-oriented work examines the generated ideas, themselves, either against human ideas [4], [11], or across models [12], [13], [22]. Read together, these studies show a

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divergence between the individual and the population level: higher individual quality ratings may accompany a decline in collective diversity [5], chatbot ideas are not always qualitatively distinct from human ones [4], comparable originality can coexist with much lower population variability [13] or with abundant but redundant output [22], and prompting strategy alone can shift the variance of the idea pool substantially [23]. Most of this work relies on human raters or LLM panels and a single creative task. Ruan et al. [12] are an exception in domain coverage but applied an LLM-as-judge instrument to English scientific domains.

B. Automatic Evaluation of Generative Outputs

Automatic metrics for generated text group according to the signal they capture: surface-level metrics measure variation through lexical overlap among a set of outputs [9], while representational metrics compare texts through embeddings designed for cosine similarity [10], [24]. Diversity is an evaluation axis in its own right with no standard measurement method [7], and taxonomies of NLG metrics together with critiques of their use have been compiled systematically [8]. The metrics also behave in ways that complicate interpretation. Transformer-based representations are anisotropic, so every comparison carries a non-zero baseline [17], cosine similarity in learned embeddings can reflect factors other than semantic proximity [18], automatic scoring on creative tasks yields a compressed score range [12], and semantic-distance approaches correspond only partially with human judgment [25]. Metrics are nonetheless frequently reported individually or aggregated without examination of their distributional behavior [8], even though lexical overlap accounts for part of the observed semantic similarity [13].

C. Indonesian and Context-Sensitive LLM Evaluation

Evaluation resources for Indonesian have developed through benchmarks with different targets. IndoNLU [14] covers understanding tasks, IndoNLG [16] extends coverage to generation through tasks with well-defined references, IndoMMLU [15] targets model knowledge, and NusaCrowd [26] consolidates Indonesian and local-language resources into a single suite. Task-specific work on Indonesian text has likewise concentrated on classification against labeled references, as in the modified encoder architecture proposed for hate speech and abusive language detection in Indonesian-English code-mixed text [27]. Creative ideation differs from all of these because no reference answer is available, so assessment must rest on the internal characteristics of the output set. Heterogeneous application domains compound the problem, because model behavior in one domain need not describe its behavior in another [12]. Across these approaches the instrument, the dimensions reported, the languages, and task scope all differ; none combines simultaneous multi-dimensional automatic characterization with an explicit examination of metric behavior in an Indonesian cross-domain ideation setting. This study performs such a characterization; it does not seek a universal winner, propose a new creativity metric, or validate conceptual creativity against human judgement.

III. Experimental Setup

A. Research Design

Fig. 1 summarizes the experimental pipeline. This study uses a controlled comparative design with two factors, the generator model and the application domain. The prompt, the number of ideas per problem statement, the temperature, and every scoring procedure were identical for both models, so observed differences can be attributed to the model rather than to treatment. The analysis is directed at characterization rather than ranking: each dimension is analyzed separately, and the composite score is used only for internal selection.

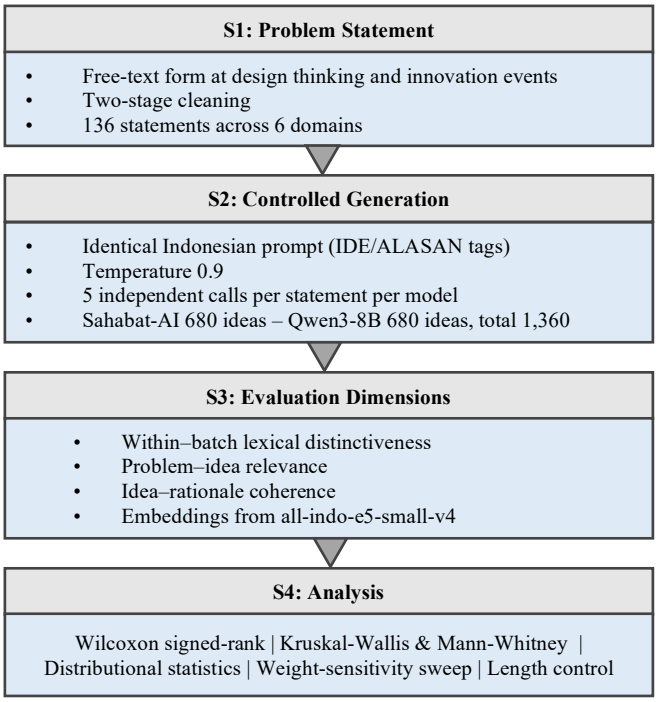


Fig. 1. Overview of the Experimental Pipeline

B. Problem Statement Dataset

The initial data consisted of everyday problem statements collected through a single free-text form administered to participants in design thinking classes, hackathons, product innovation workshops, and digital business and entrepreneurship competitions. Participation was voluntary and respondents consented to the research use of their submissions, all records were anonymized before analysis, and no personally identifying information was retained. Cleaning proceeded in two stages. Entries that did not meet the characteristics of a problem statement, namely personal narratives and complaints containing no structured problem, were removed on the basis of substring matching. The free-text sector field was then recoded into six main categories; six label strings fell outside them, among them "*Peternakan*" (livestock farming) and "*Masalah lalu lintas*" (traffic problems), and each was assigned by the authors from the content of the associated statement.

The six domains reflect the distribution of the collected problems rather than an established sectoral taxonomy. The final dataset comprises 136 problem statements, ranging from 19 to 152 words with a median of 29. The domain counts are local MSMEs and community economy 30, health and social welfare 26, education and digital literacy 25, public services in remote areas 23, environment and sustainability 18, and local food security with traditional farming and fisheries 14.

C. Models and Generation Procedure

Two LLMs of comparable scale were run locally through Ollama: Sahabat-AI [19], a continued pre-training adaptation of Llama-3 8B for Indonesian, obtained as the community-

packaged build csalab/sahabatai1 [28], and Qwen3-8B [20], a recent multilingual model. The pairing deliberately contrasts a locally adapted model with a general multilingual model at comparable parameter scale.

TABLE I. GENERATION AND SCORING CONFIGURATION

Parameter	Sahabat-AI	Qwen3-8B
Ollama model tag	csalab/sahabatai1: llama3 instruct Q4 K M	Qwen3:8b
Architecture	llama	Qwen3
Parameter count	8.0B	8.2B
Quantization	Q4 K M	Q4 K M
Model context length	8,192	40,960
Temperature	0.9 (set explicitly)	0.9 (set explicitly)
top k	40 (Ollama default)	20 (packaged)
top p	0.9 (Ollama default)	0.95 (packaged)
repeat penalty	1.1 (Ollama default)	1.0 (packaged)
Ideas per problem statement	5	5
Retries on parse failure	Up to 2	Up to 2
Embedding model	all-indo-e5-small-v4 (384-dim)	all-indo-e5-small-v4 (384-dim)
Embedding prefix / normalization	passage:/L2	passage:/L2
Encoding batch size	64	64
Minimum token length	3 characters	3 characters
Relevance weights (cosine: keyword)	0.6 : 0.4	0.6 : 0.4
Length-matching threshold	at most 2 words	at most 2 words

Each problem statement was submitted to both models with an identical Indonesian prompt requesting one creative idea and a two to three sentence explanation, in a structured format marked by the tags IDE (idea) and ALASAN (rationale). The temperature was set to 0.9 for both models, and each produced five ideas per problem statement, from independent calls, with up to two retries when parsing failed.

Table I summarizes the configuration. Only the temperature was set explicitly; the remaining decoding parameters were left at each build's packaged defaults, which are not identical, and no random seed was fixed, so the generated set can be released and reanalyzed but not regenerated identically. Section IV-E returns to this decoding asymmetry.

All 1,360 responses contained both markers, so parsing fallback was never invoked and no reasoning traces appeared in any stored response. Ideas shorter than eight words were flagged but retained, numbering 24, of which 23 came from Sahabat-AI, which also spanned a wider length range, 2 to 59 words against 3 to 38.

D. Evaluation Dimensions

Three dimensions were computed for each idea. Semantic representations were obtained with all-indo-e5-small-v4 [29], a 384-dimensional Indonesian sentence embedding model built on the E5 architecture and fine-tuned on Indonesian retrieval and question answering data. The passage: prefix was applied uniformly to problem statements, ideas, and rationales, and embeddings were L2 normalized before cosine similarity was computed [10].

1) Within-batch lexical distinctiveness

Each idea is represented as the set of unique word types after tokenization and stop word removal. Let T_i denote the word-type set of idea i and B the batch of five ideas generated by the same model for the same problem statement. Distinctiveness is defined as

$$d_i = 1 - max_{j \in B, j \neq i}\left(\frac{|Ti \cap Tj|}{|Ti \cup Tj|}\right) \quad (1)$$

The maximum detects redundancy against a single nearest idea, which a pairwise mean would mask, but is monotone in batch size, so the metric is comparable only across equal-sized batches; all 272 problem-model combinations contain exactly five ideas. The lexical approach was adopted after embedding-based measurement proved insufficiently discriminative: ideas for one problem statement occupied a very dense region of the representation space.

2) Problem-idea relevance

This dimension combines two signals additively. Let S_i be the cosine similarity between the idea embedding and the problem statement embedding, and k_i the proportion of non-stopword keywords from the problem statement that reappear in the idea text, both min-max normalized over the full set of 1,360 ideas. The composite relevance score is

$$R_i = w.S_i + (1 - w).k_i, w = 0.6 \quad (2)$$

The raw sub-scores were retained and analyzed separately, since one aim of the study is to examine whether aggregation conceals differences visible in the components.

3) Idea-rationale semantic coherence

Let e_i and r_i be the L2-normalized embeddings of idea i and of its accompanying rationale. Coherence is defined as

$$c_i = \cos(e_i, r_i) = e_i.r_i \quad (3)$$

The measure describes the semantic connection between an idea and its supporting explanation and is not claimed to measure explainability. All statistical analyses use raw scores; per-domain normalization would remove differences between domains by construction.

E. Statistical Analysis

Model differences were tested with the Wilcoxon signed-rank test at the problem statement level, each model's value being the mean of its five ideas, giving 136 matched pairs, with matched-pairs rank-biserial correlation as the effect size.

Cross-domain behavior was examined in two steps. Kruskal-Wallis screened for variation across domains within each model, with epsilon squared as the effect size. Where variation was present, Mann-Whitney was applied per domain to check whether the direction held, with Bonferroni correction over six comparisons and Cliff's delta as the effect size.

Metric behavior was examined through distributional statistics, Spearman intercorrelations, and a sweep of the weight w in (2) from 0 to 1. Output length was controlled in two ways: Spearman partial correlation with idea length as covariate, and greedy pairwise matching within each problem statement of ideas differing by at most two words, giving 367 pairs. Variance explained by length is reported as the squared Spearman correlation. Non-parametric tests were used throughout.

IV. RESULTS AND DISCUSSION

A. Model-level Differences and the Effect of Aggregation

The left panel of Table II compares the models on 136 problem-statement-level pairs. The profiles are opposed:

Sahabat-AI leads on lexical distinctiveness, Qwen3-8B on cosine relevance and idea-rationale coherence, all with large effect sizes. Relevance is the informative case. Its two sub-components move in opposite directions with effect sizes above 0.53, and the composite combining them shows no significant difference (p=0.067): two large opposing effects cancel under aggregation.

The cancellation is a property of the weighting, not of the models. Fig. 2 sweeps the cosine weight w of (2) from 0 to 1: the conclusion reverses between 0.70 and 0.75, is significant with a large effect at both ends but points to opposite models, and is absent in between, all from the same data. The value w=0.6 used here falls in the transition zone, so its null result is a finding about aggregation rather than about the models.

TABLE II. MODEL COMPARISON BEFORE AND AFTER CONTROLLING FOR OUTPUT LENGTH

Dimension	Sahabat-AI	Qwen3-8B	p	Rank-bis.	P (matched)	Rank-bis. (matched)	Δ|r|
Lexical distinctiveness	0.738	0.663	2.06x10-9	+0.592	1.5x10-11	+0.409	-0.183
Relevance (cosine)	0.912	0.932	5.04x10-8	-0.539	0.864	-0.012	-0.527
Relevance (keyword overlap)	0.175	0.142	7.25x10-9	+0.580	5.3x10-17	+0.565	-0.016
Relevance (composite)	0.647	0.642	0.067	+0.181	4.4x10-17	+0.522	+0.341
Idea-rationale coherence	0.938	0.966	2.58x10-11	-0.659	3.1x10-3	-0.179	-0.480

Left panel: Wilcoxon signed-rank on 136 problem-statement-level pairs. Right panel: the same test on 367 length-matched idea pairs (mean length 19.79 words for both models), group means for the matched subset differ from the full sample by at most 0.009. Δ|r| is the change in the absolute rank-biserial correlation between the two panels, the direction of every effect is unchanged.

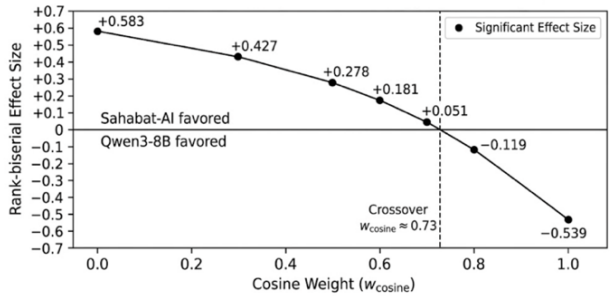


Fig. 2. Matched-pairs rank-biserial correlation versus cosine weight.

B. Cross-Domain Consistency

Kruskal-Wallis screening indicates cross-domain variation in nine of ten dimension-model combinations. The exception is idea-rationale coherence for Sahabat-AI (H=2.90, p=0.715, ε^2=0.004), stable across domains while the same dimension varies for Qwen3-8B (H=23.78, p<0.001, ε^2=0.035). Across the nine significant combinations, ε^2 ranges from 0.019 to 0.048. Sensitivity to domain is thus itself a model characteristic.



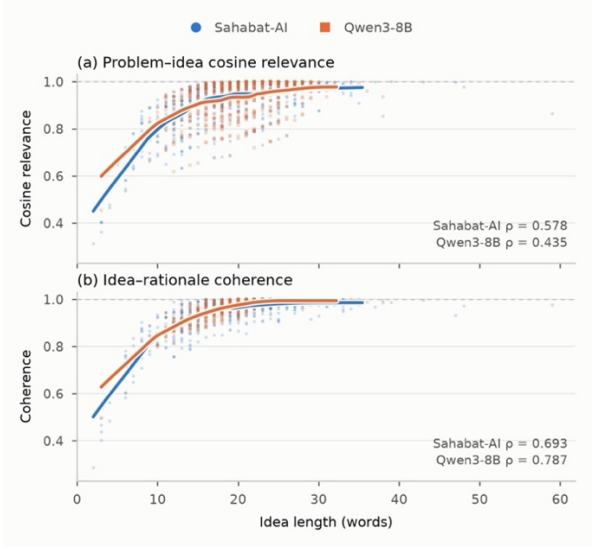
Positive values favor Sahabat-AI and negative values Qwen3-8B, asterisks mark comparisons that remain significant after Bonferroni correction over the six domains

Fig. 3. Cliff's delta for the model contrast in each dimension and domain.

Directional consistency is dimension-dependent, as Fig. 3 shows. Lexical distinctiveness is the most stable, holding its direction in all six domains, surviving Bonferroni correction in five, and carrying the largest effects observed (Cliff's delta 0.154 to 0.539). Keyword overlap and idea-rationale coherence also hold direction in all six domains but survive correction in three each. Cosine relevance holds direction in five domains and survives in only one, and the composite score is inconsistent in both direction and significance, favoring Sahabat-AI in four domains with none significant. The figure is read by row: a dimension either keeps one sign across the row or it does not, and the composite row is the one that does not.

C. Metric Behavior and Surface Variables

This subsection turns from the models to the instruments used to compare them. Both embedding-based measures concentrate near the ceiling. Idea-rationale coherence has a median of 0.975, an interquartile range of only 0.057, a skewness of -3.74, and 31.7% of its values above 0.99, while cosine relevance has a median of 0.959, a skewness of -2.02, and 24.6% above 0.99. The two lexical measures are spread far more widely and produce no value above that threshold. This fits the anisotropy of transformer representations and the evidence that cosine similarity reflects factors beyond semantic proximity.



Each point is one of the 1,360 ideas, curves are per-model monotone local regression fits, and Spearman ρ is reported in each panel. The horizontal range is truncated at the 99th percentile of idea length.

Fig. 4. Idea length against the two embedding-based measures: (a) problem-idea cosine relevance, (b) idea-rationale coherence.

Output length is one such factor. Fig. 4 plots it against both embedding-based measures. The relationship is steep and

saturating rather than linear: scores rise sharply up to roughly fifteen words, then compress against the ceiling. Length explains under 0.6% of the variance in lexical distinctiveness but 62.0% and 48.1% of coherence for Qwen3-8B and Sahabat-AI, and 18.9% and 33.4% of cosine relevance. The separation between the two models is concentrated at short outputs: the curves stand apart below roughly twelve words, where 23 of the 24 ideas shorter than eight words come from Sahabat-AI, and they converge thereafter. Once length is equalized over 367 matched pairs, in the right panel of Table II, the cosine relevance difference disappears ($\rho = -0.021$, $p = 0.45$) and coherence loses about 70% of its effect, so a difference significant at 10-8 traces partly to a surface variable. The composite behaves inversely, non-significant in the full sample but significant with a large effect once length is equalized, so its earlier null combined two distortions. That the two sub-components labeled relevance correlate at only 0.481 makes the same point: a shared label does not guarantee a shared construct.

D. Qualitative Illustration

Table III shows one batch from the education domain, for the problem "*Banyak siswa di desa terpencil kesulitan mengakses buku bacaan dan bahan belajar tambahan karena perpustakaan terbatas dan toko buku tidak tersedia di sekitar mereka*". Three ideas from each model are reproduced; all generation and scoring were carried out on the Indonesian text.

The batch exhibits all three findings at once. Qwen3-8B keeps to a single frame, opening its three ideas with the same construction, and two are close enough to be interchangeable (distinctiveness 0.545) while each tracks its rationale almost exactly (up to 0.999). Sahabat-AI ranges more widely, but its third idea consists of only two words: it scores 0.857 on lexical distinctiveness and collapses on both embedding measures (0.313 and 0.287), because a two-word text has a small word-type set and a weak semantic signal at once. The dimensions disagree on the same output, systematically rather than as noise.

TABLE III. EXAMPLES OF GENERATED IDEAS FROM ONE BATCH

Model	Generated idea	Dist.	Cos.	KW	Coh.
Sahabat-AI	Membangun perpustakaan daring dengan koleksi buku digital dan ruang belajar online yang dapat diakses melalui internet.	0.778	0.970	0.188	0.960
Sahabat-AI	Membangun "BiblioBox", yaitu sebuah kotak kecil yang dapat menampung beberapa buku bacaan dan peralatan belajar lainnya, yang akan disimpan di tempat – tempat strategis seperti pos kesehatan desa …	0.903	0.987	0.250	0.974
Sahabat-AI	Perpustakaan keliling.	0.857	0.313	0.062	0.287
Qwen3-8B	Membuat platform digital berbasis komunitas yang menyediakan buku bacaan dan bahan belajar dalam format digital yang bisa diakses melalui smartphone atau komputer dengan koneksi internet terbatas.	0.545	0.997	0.312	0.999
Qwen3-8B	Membuat "Perpustakaan Gerak Digital" berupa mobil yang dilengkapi e-reader dan koneksi internet, bergerak ke desa terpencil …	0.800	0.997	0.250	0.990
Qwen3-8B	Membuat perpustakaan digital berbasis aplikasi dengan konten yang dapat diunduh dan diakses melalui smartphone atau komputer dengan koneksi terbatas.	0.545	0.990	0.125	0.921

E. Implications and Limitations

These results argue for reporting dimensions separately, as design research has long done for ideation [21], including distributional diagnostics, testing sensitivity to aggregation parameters, and controlling surface variables before capability differences are inferred, otherwise an artifact reads as capability, or a real difference stays masked. For Indonesian, a locally adapted and a general multilingual model produce systematically different profiles, so model selection should follow the dimensions the application requires rather than a single ranking.

The two models were not fully symmetric in their decoding configuration. A narrower top_k and a lower repeat_penalty plausibly reduce lexical variation, and Sahabat-AI also produced 23 of the 24 ideas shorter than eight words, whose small word-type sets inflate Jaccard-based distinctiveness. Part of the gap in Section IV-A may therefore reflect decoding defaults rather than model behavior, and re-running under harmonized parameters is a necessary next step.

Further limits apply. Two models in one setting cannot generalize to all LLMs, and the dataset was gathered through a single instrument from a population already practiced in formulating problem statements, so it does not represent the general Indonesian context. The three dimensions are operational constructs, expressing neither conceptual novelty, human-perceived relevance, nor explainability. Length control does not exclude other unmeasured surface variables, and no validation against human judgment was performed, which matters because semantic-distance measures agree only partially with human scoring [25].

V. CONCLUSION

This study asked what a model comparison on open-ended creative output actually rests on. Using 136 Indonesian design thinking problem statements from six application domains and 1,360 ideas generated by two LLMs of comparable scale, it evaluated three dimensions simultaneously and then subjected the resulting comparison to three checks.

Varying the aggregation weight reverses the direction of the conclusion, so the same data supports opposite claims depending on a weighting for which the literature provides no validated empirical basis. The present study therefore reports the full weight sweep rather than committing to a single composite score. The embedding-based measures concentrate near the ceiling, which limits how far they can discriminate at all. Equalizing output length removes the cosine-based relevance difference entirely and reduces the coherence difference by about seventy percent. Two sub-components that both carry the label relevance correlate at only 0.481, so a shared name does not guarantee a shared construct.

Against that background, the model-level result is stated with the qualification it requires. The two systems display opposing profiles rather than a better-or-worse relationship: Sahabat-AI produces ideas that are more lexically distinct within a batch, Qwen3-8B output with higher semantic proximity and idea-rationale coherence. Only the lexical

advantage survives length control with moderate attenuation, and consistency across domains proves conditional on the dimension observed.

The contribution is therefore methodological before it is comparative: an account of how aggregation choices and surface variables can manufacture or erase an apparent capability gap, demonstrated on a controlled cross-domain Indonesian setting and released with the code that reproduces every reported statistic. Three directions follow: validation against expert judgment to establish what the three dimensions measure, extension to more models and domains to test the generality of the effects, and evaluation procedures that control surface variables by construction rather than after the fact.

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